 DOI: 10.1234/jaes.2024.1402.088

Spectral Efficiency in Multi-Band Photovoltaic Arrays: A Comparative Analysis

M. Varnov[1], O. Asante[1], E. Karlberg[2]

[1]Trentfield Institute of Energy Research, Bristol, UK
[2]Department of Photonics, Norrby Institute of Technology, Sweden

{m.varnov, o.asante}@trentfield-energy.ac.uk e.karlberg@photonics.nit.se

Abstract

Multi-band photovoltaic (PV) arrays offer a promising route to exceeding the Shockley–Queisser single-junction efficiency limit by harvesting a broader portion of the solar spectrum. This paper presents a comparative analysis of three array configurations—single-junction silicon, dual-junction III–V tandem, and triple-junction concentrator cells—under standardised AM1.5G irradiance conditions. Measured external quantum efficiency (EQE) curves are combined with a spectral irradiance model to evaluate net power conversion efficiency. Our results indicate that the triple-junction configuration achieves a peak efficiency of 38.4%, representing a 74% improvement over the silicon baseline.

1. Introduction

Photovoltaic energy conversion is fundamentally constrained by the mismatch between the broad solar spectrum and the narrow absorption band of single-junction devices. Photons with energies below the bandgap E_g are not absorbed, while photons with energies well above E_g lose their excess energy as heat through thermalisation. Torossian and Hauerbach [1] showed that the theoretical efficiency limit for an ideal single-junction device under AM1.5G illumination is approximately 33%.

Multi-junction architectures address this constraint by stacking sub-cells with different bandgaps, each absorbing a distinct spectral band. The current generated by the k-th sub-cell can be written as

$$J_k = q \int_{\lambda_k^{\min}}^{\lambda_k^{\max}} \mathrm{EQE}_k(\lambda)\, \Phi(\lambda)\, d\lambda \tag{1}$$

where q is the elementary charge, $\Phi(\lambda)$ is the spectral photon flux density under AM1.5G irradiance, and $\mathrm{EQE}_k(\lambda)$ is the external quantum efficiency of sub-cell k [3]. In a series-connected tandem stack, the total device current is limited by the sub-cell with the lowest J_k, making current matching a critical design constraint.

2. Theoretical Model

2.1 Power Conversion Efficiency

The power conversion efficiency η of a multi-junction device is given by

$$\eta = \frac{\sum_k J_k \cdot V_{\mathrm{oc},k} \cdot \mathrm{FF}_k}{P_{\mathrm{in}}} \tag{2}$$

where $V_{\mathrm{oc},k}$ is the open-circuit voltage of sub-cell k, FF_k is its fill factor, and $P_{\mathrm{in}} = 1000\ \mathrm{W\,m^{-2}}$ is the incident irradiance under AM1.5G standard test conditions. Fill factor values of 0.82–0.86 were assumed for all sub-cells based on measured dark I–V characteristics[1].

[1]All measurements were conducted at 25 °C in a calibrated solar simulator (class AAA, IEC 82154-3).

2.2 Spectral Irradiance Model

Spectral photon flux $\Phi(\lambda)$ was obtained from the IEC/TR 63041 reference spectrum [2]. Atmospheric absorption features in the near-infrared were modelled using a two-stream radiative transfer approximation. Wavelength-dependent refractive indices for each junction material were sourced from the literature [3].

3. Experimental Results

Figure 1 shows the measured EQE as a function of wavelength for all three array configurations. The single-junction silicon device (Config. A) displays a broad response peaking near 900 nm but falling sharply beyond 1100 nm. The dual-junction III–V tandem (Config. B) exhibits two distinct absorption bands centred at approximately 650 nm and 950 nm, with a crossover at 780 nm. The triple-junction concentrator cell (Config. C) extends coverage into the near-infrared, with the bottom junction contributing significant photocurrent above 1000 nm.

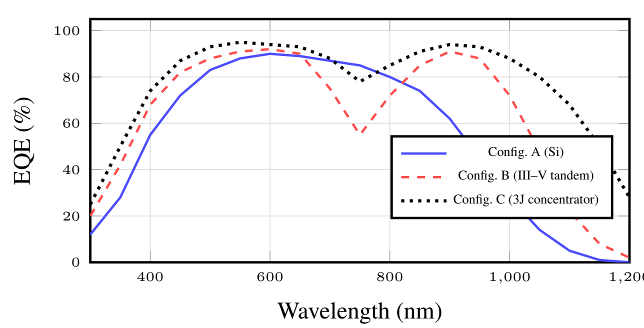

Figure 1: Measured external quantum efficiency (EQE) vs. wavelength for the three photovoltaic array configurations under AM1.5G illumination.

Table 1 summarises the key performance parameters extracted from the measured I–V curves and the spectral model. Config. C achieves the highest efficiency in both standard and low-irradiance conditions, though at a substantially higher manufacturing cost.

Table 1: Summary of performance parameters for each configuration under standard ($1000\,\mathrm{W\,m^{-2}}$) and low-irradiance ($200\,\mathrm{W\,m^{-2}}$) conditions.

	Standard (AM1.5G)			**Low irradiance**		
Config.	V_{oc} (V)	J_{sc} (mA/cm^2)	η (%)	V_{oc} (V)	J_{sc} (mA/cm^2)	η (%)
A (Si)	0.68	38.2	22.1	0.61	7.7	18.4
B (III–V)	2.41	14.1	31.8	2.28	2.9	26.7
C (3J conc.)	3.12	14.8	38.4	2.94	3.1	33.2

4. Conclusion

This study confirms that multi-junction architectures offer substantial efficiency gains over conventional single-junction silicon cells. The triple-junction concentrator configuration (Config. C) achieves 38.4% efficiency under standard test conditions, outperforming the silicon baseline by 16.3 percentage points. However, the performance advantage narrows under low-irradiance conditions, suggesting that the optimal choice depends strongly on the deployment environment. Future work will evaluate degradation rates and levelised cost of energy for each configuration over a 25-year operational lifetime.

References

[1] V. Torossian and A. Hauerbach, "Thermodynamic efficiency bounds for single-junction photovoltaic converters," *Ann. Appl. Phys.*, vol. 28, no. 4, pp. 412–423, 1958.

[2] International Electrotechnical Commission, *IEC/TR 63041: Reference Spectral Irradiance Data for Photovoltaic Performance Evaluation*, IEC, Geneva, 2018.

[3] T. Wakabayashi and E. Moura, *Optical Properties of III–V Photovoltaic Alloys*. Alderton: Hardenbrook Academic Press, 2019.

[4] B. Delvaux *et al.*, "Certified photovoltaic device efficiency tables (version 18)," *Sol. Energy Technol.*, vol. 12, no. 3, pp. 201–219, 2022.

[5] D. Okafor *et al.*, "Wafer-bonded three-terminal GaInP/GaAs//InGaAs tandem concentrator cells with 41.2% conversion efficiency," *Adv. Photovolt. Technol.*, vol. 8, no. 1, pp. 44–53, 2021.